

Muthike Ajay

+1 (302) 266 1123 | ajaymuthike1@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data Scientist with 4+ years of experience in designing and deploying scalable data pipelines and machine learning solutions. Expertise in Python-based data preparation, transformation, and exploratory analysis using Databricks, PySpark, Apache NiFi, and Airflow, complemented by Azure and AWS cloud deployments. Developed real-time analytics platforms and LSTM forecasting models, delivering actionable insights and optimizing operational performance.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL
- **Data Engineering & Big Data:** Apache Spark, Apache Kafka, Databricks, Apache NiFi, Airflow, Hadoop, HBase
- **Machine Learning & AI:** Scikit-Learn, TensorFlow, Keras, PyTorch, Hugging Face
- **Data Analysis & Statistics:** SAS, SPSS, Bayesian Analysis, Multivariate Analysis, A/B Testing, Monte Carlo Simulations, LSTM Networks, Time Series Forecasting, Portfolio Optimization, Data Analysis, Analytical Skills
- **Data Visualization:** Power BI, Tableau, Plotly
- **Model Deployment & MLOps:** Docker, Kubernetes, FastAPI, MLflow
- **Cloud Platforms:** AWS, Azure
- **Data Storage & Warehousing:** AWS S3, Amazon Redshift, SQL Databases
- **APIs & Web Services:** RESTful APIs, FastAPI, Flask
- **Compliance & Security:** HIPAA, FHIR, HL7, ICD, RBI, SEBI
- **Project Management & Version Control:** Git, GitHub, Jira
- **Operating Systems:** Windows, Linux, Mac Ios

PROFESSIONAL EXPERIENCE

American Express-DE | *Data Scientist*

May 2024 - Present

- Built a real-time fraud analytics platform using Python, Apache Kafka, and PySpark, handling millions of transactions per day and achieving 99% accuracy in fraud detection while demonstrating strong data analysis skills.
- Designed efficient ETL workflows with Apache NiFi and Airflow, consolidating multi-source data and cutting processing time by 40%.
- Developed dynamic Tableau dashboards to monitor critical KPIs, empowering leadership with data-driven insights and enhancing reporting speed by 35% through rigorous analytical skills.
- Applied NLP methods to evaluate customer reviews and social media sentiment, contributing to a 20% uplift in customer satisfaction metrics.
- Built a comprehensive A/B testing mechanism for evaluating new product releases, leading to a 12% increase in feature adoption and enabling data-driven decision making.
- Designed a time-series prediction model using LSTM architectures for market forecasting, improving investment decision accuracy by 18%.
- Established a Docker- and Kubernetes-based ML deployment pipeline, decreasing production rollout time from 10 hours to 1 hour and streamlining model operationalization.
- Enhanced data storage and querying using Hadoop and HBase, improving performance and reducing retrieval latency by 60% across massive datasets.
- Used advanced portfolio optimization through Monte Carlo simulations, delivering a 10% improvement in risk-adjusted returns for premium clients.
- Built and deployed a real-time credit scoring API on Azure using Flask, facilitating instant loan risk evaluation.
- Leveraged distributed computing with Apache Spark and Databricks to transform and analyze historical transaction records, revealing previously unknown behaviors and trends.

Zensar Technologies – India | *Data Scientist*

Jan 2021 - Jul 2023

- Fine-tuned a Large Language Model (LLM) on AWS Bedrock to enhance predictive analytics for healthcare risk assessment and patient readmission forecasting, resulting in an 18% reduction in readmission rates and improved care management outcomes.
- Developed a highly efficient semantic search system for clinical notes using Retrieval-Augmented Generation (RAG), leveraging fine-tuned LLM embeddings and Hugging Face cross-encoders for intelligent re-ranking, enabling clinicians to quickly extract critical patient context for faster and more accurate diagnoses.
- Utilized Apache Spark and Databricks for large-scale distributed processing of healthcare data, improving compute efficiency and reducing ETL pipeline duration by 40% while ensuring support for complex analytics workloads.
- Designed and automated end-to-end ETL workflows with Apache NiFi and Talend to seamlessly integrate EHR/EMR records, increasing data freshness, accuracy, and interoperability across multiple clinical systems by 25%.
- Maintained strict data security posture by implementing HIPAA-compliant encryption, anonymization, access controls, and audit mechanisms to protect patient health information (PHI).
- Leveraged AWS S3 for scalable storage and Amazon Redshift for high-performance warehousing, ensuring 99% availability and reliable analytics access for cross-functional teams.
- Built actionable insights dashboards using Power BI and Plotly, empowering physicians and healthcare leaders with real-time views of patient risk profiles, operational trends, and intervention effectiveness through strong analytical skills.
- Ensured data standardization through compliance with leading formats such as FHIR, HL7, and ICD, improving data quality and cross-system communication.
- Applied advanced Bayesian inference and multivariate statistical techniques using SAS and SPSS to support research studies, validate treatment outcomes, and drive evidence-based medical decisions.

EDUCATION

Wilmington University

Masters of Information System Technologies, Information System Technologies

New Castle, Delaware, USA

Vignan University

Bachelor of Technology

Vadlamudi, Guntur, India